2023-2027/CSE/2/1

NIRNAY - X

A Project Report

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Technology in Computer Science and Engineering

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August, 2026

Declaration

We hereby declare that this project report entitled “Nirnay-X ” submitted to the Department of Computer Science and Engineering, **Ajay Kumar Garg Engineering College**, **Ghaziabad** in partial fulfillment of the requirements for the degree of Bachelor of Technology is an authentic record of our original work carried out during the academic year 2023-2027.

The work contained in this report has not been submitted in part or in full to any other university or institution for the award of any degree or diploma. All sources of information used in this report have been duly acknowledged and referenced.

We have self-checked this report for plagiarism using appropriate tools and confirm that it meets the prescribed academic integrity standards.

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Certificate

This is to certify that the project report entitled “NIRNAY-X" submitted by the following students in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering to **Ajay Kumar Garg Engineering College**, Ghaziabad (affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow) is an authentic record of genuine work carried out by them under my supervision.

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The matter presented in this report has not been submitted for the award of any other degree or diploma at this or any other institution.

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Acknowledgement

We express our deepest gratitude to our project supervisor, **Ms. Bhumica**, Department of Computer Science & Engineering, Ajay Kumar Garg Engineering College, Ghaziabad, for her invaluable guidance, constant encouragement, and constructive feedback throughout the course of this project. Their expertise and insights have been instrumental in shaping this work.

We would like to extend our sincere thanks to **Dr. Amita Dev**, Advisor, for her valuable suggestions, guidance, and support, which greatly contributed to the successful completion of this project. We are highly grateful to **Dr. Hemant Ahuja**, Director, **Ajay Kumar Garg Engineering College, Ghaziabad**, for providing institutional support and fostering an environment that encourages innovation and research.

We sincerely thank Dr. Anu Chaudhary, Head of the Department of Computer Science & Engineering, for providing the necessary infrastructure, lab facilities, and an environment conducive to research and learning.

We extend our heartfelt thanks to the faculty members of the Department of Computer Science & Engineering for their continuous support and for imparting the foundational knowledge that has made this project possible.

We are grateful to the management of **AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD**, for maintaining an ecosystem that encourages innovation and applied research.

Finally, we acknowledge our families and friends for their unwavering moral support and patience during the entire duration of this project.

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Abstract

Nirnay-X is an intelligent legal assistance platform designed to bridge the gap between complex legal systems and the people who need them. The platform empowers both legal professionals and common citizens by providing accurate, contextual, and easy-to-understand legal guidance using Artificial Intelligence, retrieval-augmented generation (RAG), and large language models (LLMs).

The system allows users to upload legal documents such as contracts, FIRs, court notices, judgments, and petitions in PDF format. These documents are converted into vector embeddings and stored in a secure vector database for semantic retrieval. When a user asks a legal question, the AI retrieves the most relevant legal information from the uploaded documents as well as an indexed constitutional and statutory knowledge base, ensuring context-aware and reliable responses.

For common citizens, Nirnay-X simplifies legal language into understandable explanations, identifies applicable sections and subsections of the law, suggests possible legal remedies, and recommends the next steps without replacing professional legal advice.

For lawyers, the platform acts as an intelligent legal assistant by analyzing case documents, retrieving relevant precedents, summarizing judgments, suggesting related case studies, tracking hearing dates, managing case timelines through integrated calendar tools, and maintaining organized client records. AI-powered tool calling automates repetitive legal workflows, allowing lawyers to focus more on strategy and client advocacy.

The platform combines document intelligence, legal research, workflow automation, and conversational AI into a single ecosystem, making legal services more accessible, efficient, and transparent.

Contents

Declaration
Certification
Acknowledgement
Abstract
1. Introduction 9
1.1. Background and motivation 9
1.1.1. Domain overview 9
1.1.2. Motivating Scenario 9
1.1.3. Problem Statement 10
1.2. Research Questions and Objectives 11
1.2.1. Primary Research Question 11
1.2.2. Secondary Research Question 11
1.2.3. Research objective 11
1.3. Novelty and Originality 12
1.4. Scope of the Project 12
1.4.1. Assumptions and Constraints 13
1.5. Organization of the Report 14
2. Literature Review 15
2.1. Review Methodology 15
2.1.1. Search Strategy and Databases 15
2.1.2. Inclusion and Exclusion Criteria 15
2.2. Historical Overview 16
2.3. Thematic Review of Current State-of-the-Art 16
2.3.1. AI and NLP for Legal Information Processing 16
2.3.2. LLMs and Retrieval-Augmented Generation 17
2.3.3. Sticeman Search, Case Retrieval and Workflow Automation 17
2.4. Comparative Summary of Reviewed Works 18
2.5. Identification of Research Gap 19
2.6. Formulation of the Research Proposal 19
3. Societal impact and SDG alignment 21
3.1. Primary SDG relevance 21
3.1.1. Mapped SDG and Target 21

3.1.2. Alignment Justification 21
3.2. Breadth of SDG linkage 22
3.2.1. Secondary SDG 1: SDG 3 – Good Health and Well-being 22
3.2.2. Secondary SDG 2: SDG 9 – Industry, Innovation and Infrastructure
3.3. Stakeholder analysis 23
4. Universal human values & harmony with nature 25
4.1. Impact on material, plant/bio order 25
4.1.1. Material order 25
4.1.2. Plant/bio order 25
4.2. Impact on animal and human order 26
4.2.1. Animal order 26
4.2.2. Human order 26
4.3. Overall mutual fulfillment 27
5. Research methodology 28
5.1. Research approach 28
5.1.1. Justification of choosen approach 28
5.2. Support system/Algorithm/Framework 28
5.2.1. System architecture/perceptual model 28
5.2.2. Algorithmic/Design Details 29
5.2.3. Novelty of the proposed approach 29
5.3. Data collection and analysis 29
5.3.1. Data sources/Dataset description 29
5.3.2. Analysis techniques 30
5.4. Experimental design and validation plan 30
5.4.1. Evaluation matrices 31
5.4.2. Baseline/ Benchmark comparison 31
5.5. Tools, Technology, Resources 31
6. Expected outcomes and dissemination plan 33
6.1. Expected research contributions 33
6.2. Mode of dissemination 34
6.2.1. Conference publication 34
6.2.2. Journal publication 34
6.2.3. Patent potential 34
6.3. Outcome mapping table 35

6.4. Anticipated challenges to dissemination 36
7. Project plan and professional practices 37
7.1. Timeline and milestone 38
7.1.1. Work breakdown structure 38
7.2. Collaboration and documentation 39
7.2.1. Team roles and responsibilities 39
7.2.2. Collaborative tools and version control 39
8. Bibliography 41

1. Introduction

1.1 Background and Motivation

1.1.1 Domain Overview

The legal domain involves a large body of information, including constitutional provisions, statutes, judicial precedents, regulations, contracts, petitions, notices, judgments, and other legal documents. Accessing and interpreting this information can require familiarity with legal terminology, procedural rules, and the context in which a particular legal provision applies.

Digital legal databases have improved access to statutes and judicial decisions, but users may still need to manually search, interpret, compare, and organize the retrieved information. Legal professionals may also need to review large volumes of documents while preparing cases and tracking hearings, deadlines, and case records.

Recent developments in Artificial Intelligence (AI), Natural Language Processing (NLP), Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG) provide technical approaches for assisting with these activities. RAG combines information retrieval with language generation by retrieving relevant information from an external knowledge base before generating a response. Semantic search using vector embeddings can also retrieve information based on contextual similarity rather than exact keyword matching.

Nirnay-X applies these technologies to legal document analysis and information retrieval. The system is intended to assist users in understanding legal documents and retrieving relevant legal information while keeping professional legal judgment outside the system's role.

1.1.2 Motivating Scenario

A user may possess a legal document such as a contract, FIR, court notice, petition, or judgment but may have difficulty identifying the important information contained within it. Similarly, a lawyer preparing a case may need to examine

documents, search for relevant legal provisions, identify related judgments, and organize case information.

Nirnay-X is designed to support these activities through document processing, semantic retrieval, RAG-based question answering, document summarization, legal information retrieval, and case-management features. Users can upload legal documents, ask questions in natural language, and receive responses based on information retrieved from the available legal sources. The system is intended as an information and decision-support tool rather than a replacement for advocates or courts.

1.2 Problem Statement

Legal information is extensive and often contains specialized terminology. Users without formal legal training may have difficulty understanding legal documents, identifying potentially relevant provisions, and locating appropriate procedural information. Legal professionals may also spend considerable time reviewing documents, searching for precedents, and organizing case-related information.

Conventional keyword-based search can be useful when the appropriate terminology is known, but semantic relationships between legal concepts may not always be captured through exact word matching. At the same time, general-purpose LLMs can generate inaccurate or unsupported information when they do not have access to reliable external sources.

Therefore, the project addresses the problem of developing a unified system that can:

- Process and analyze legal documents.
- Retrieve relevant legal provisions and judicial information using semantic search.
- Generate responses using retrieved legal context through RAG.
- Explain complex legal information in simpler language.
- Support legal document organization and case-related workflows.
- Associate generated information with underlying sources where possible.

The system does not provide legally binding advice or replace professional legal judgment.

1.3 Research Questions and Objectives

1.3.1 Primary Research Question

How can AI, semantic search, and retrieval-augmented generation be integrated into a legal assistance platform to support legal document understanding and information retrieval for lawyers and common citizens?

1.3.2 Secondary Research Questions

1. How can legal documents be processed and represented for semantic retrieval?
2. How can RAG be used to provide responses based on retrieved legal information?
3. How can legal information be presented in a form that is easier for non-specialist users to understand?
4. How can document retrieval, legal research, and case-management functions be integrated into a single platform?
5. How can the reliability and limitations of AI-generated legal responses be evaluated?

1.3.3 Research Objectives

The primary objective is to design and develop Nirnay-X as an AI-powered legal intelligence platform integrating document processing, semantic search, RAG, and workflow-management capabilities.

The specific objectives are to:

- Process legal documents in PDF format.
- Extract and preprocess textual information from documents.
- Generate vector embeddings and store them in a vector database.
- Retrieve relevant legal information using semantic search.
- Generate context-aware responses using RAG and LLMs.

- Summarize lengthy legal documents.
- Explain legal terminology and concepts in simpler language.
- Retrieve potentially relevant statutory provisions and judicial precedents.
- Support case organization, hearing schedules, and related workflow activities.
- Provide a conversational interface for natural-language interaction.

The report also defines evaluation around document processing, retrieval quality, generated response quality, system performance, and functional reliability rather than relying on a single measure.

1.4 Novelty and Originality

The originality of Nirnay-X does not depend on introducing a new AI algorithm. Instead, the project focuses on integrating established technologies into a single legal-assistance platform.

The proposed system combines document processing, vector-based semantic retrieval, RAG, legal information retrieval, conversational interaction, and workflow-management features. It is designed to support both legal professionals and members of the public within the same system.

A further distinguishing feature is the combination of information retrieval with case-management functionality such as document organization, hearing schedules, and deadline tracking. The system therefore treats legal assistance as a combination of information access and workflow support rather than only a question-answering task.

1.5 Scope of the Project

The project covers the development of a web-based AI legal assistance platform capable of processing legal documents, generating embeddings, performing semantic retrieval, and producing RAG-based responses. It also includes document summarization, simplified legal explanations, legal information retrieval, case organization, and calendar-based workflow support.

The system is intended for two broad user groups: legal professionals and common citizens. For citizens, the focus is on improving understanding of legal documents and concepts. For legal professionals, the focus is on supporting research, document analysis, case organization, and related administrative activities.

The current implementation is limited to information assistance and decision support. It does not provide legally binding advice, replace advocates or judges, or independently determine court outcomes. Future extensions identified in the report include multilingual assistance, integration with official legal databases and e-Courts services, AI-assisted drafting, voice interaction, mobile support, and enhanced security.

1.5.1 Assumptions and Constraints

The system assumes that the legal information available to the retrieval system is sufficiently relevant, accurate, and current for the intended use. The quality of generated responses also depends on the quality and coverage of the underlying documents.

Important constraints include:

- Legal information may change through amendments and new judicial decisions.
- Retrieved information may be incomplete if relevant sources are absent from the knowledge base.
- AI-generated responses may contain errors and therefore require verification.
- Uploaded legal documents may contain confidential or personally identifiable information and require appropriate security controls.
- The system is not intended to replace qualified legal professionals or judicial authorities.

These constraints are particularly relevant because the report identifies the quality, completeness, and currency of the underlying legal database as factors affecting RAG-based responses.

1.6 Organization of the Report

The report is organized into six chapters. **Chapter 1** introduces the legal domain, problem statement, research questions, objectives, novelty, scope, assumptions, and constraints.

Chapter 2 presents the literature review covering LegalTech, AI, NLP, LLMs, RAG, semantic search, vector databases, legal case retrieval, and related research gaps.

Chapter 3 describes the research methodology, system architecture, data processing workflow, retrieval approach, evaluation metrics, and validation strategy.

Chapter 4 presents the implementation of Nirnay-X, including its major software modules and system functionality.

Chapter 5 discusses the system results, limitations, and possible future enhancements.

Chapter 6 concludes the report by summarizing the work and assessing the extent to which the stated objectives were addressed.

2 Literature Review

2.1 Review Methodology

The literature review examines research and technological developments related to Artificial Intelligence (AI) in the legal domain, with particular emphasis on Natural Language Processing (NLP), Large Language Models (LLMs), semantic search, Retrieval-Augmented Generation (RAG), legal document analysis, and legal case retrieval. The review is used to understand existing approaches, their capabilities and limitations, and the research gap addressed by Nirnay-X.

2.1.1 Search Strategy and Databases

The review focuses on research concerning AI-assisted legal research, legal document processing, legal question answering, judgment summarization, semantic retrieval, vector databases, RAG, and LegalTech workflow automation. The existing report discusses research and technological developments in these areas, but it does not specify a complete list of databases or search strings used for the review. Therefore, specific database names and search-query statistics are not stated here.

2.1.2 Inclusion and Exclusion Criteria

The review considers work relevant to AI-based legal information retrieval, document analysis, legal question answering, case retrieval, summarization, semantic search, and workflow automation. Approaches that contribute to understanding the use of AI for legal information processing are considered relevant.

The review focuses on technologies and approaches directly related to the objectives of Nirnay-X rather than general AI applications unrelated to legal information management.

2.2 Historical Overview

Legal research has traditionally involved the examination of statutes, judicial precedents, legal commentaries, and official records. With the digitization of legal information, computerized legal databases and keyword-based search systems enabled electronic access to legal documents. Online legal databases and e-Governance initiatives further expanded access to statutes and judicial decisions.

The development of Machine Learning and NLP introduced automated document classification, entity extraction, summarization, and legal information retrieval. More recently, transformer-based language models and LLMs have enabled conversational interaction with legal documents.

RAG introduced another important development by combining information retrieval with language generation. Instead of relying only on information encoded in a language model, RAG retrieves relevant documents from an external knowledge base and provides them as context for response generation. Semantic search and vector databases further support retrieval based on contextual similarity rather than exact keyword matching.

2.3 Thematic Review of Current State-of-the-Art

2.3.1 Theme 1: AI and NLP for Legal Information Processing

AI and NLP techniques have been applied to several legal tasks, including document classification, named entity recognition, contract analysis, judgment summarization, legal question answering, and legal knowledge extraction.

These approaches can reduce manual processing for specific tasks, but their performance depends on the task, data quality, and domain characteristics. Legal applications also require attention to jurisdiction, legal context, accuracy, and explainability. The report therefore treats AI as an assistance mechanism rather than a replacement for legal expertise.

2.3.2 Theme 2: LLMs and Retrieval-Augmented Generation

LLMs provide natural-language interaction, document summarization, and explanatory capabilities for legal applications. However, responses generated without access to reliable external information may contain inaccurate citations, omit relevant context, or produce unsupported statements.

RAG addresses part of this problem by retrieving relevant information from an external knowledge base before generating a response. This allows the retrieved material to be used as contextual evidence and can make the response more traceable. However, RAG does not guarantee correctness. Its performance depends on the relevance, completeness, and currency of the retrieved legal information.

For Nirnay-X, this approach forms the basis of the legal question-answering pipeline. Uploaded documents are processed, divided into chunks, converted into embeddings, stored in a vector database, and retrieved when a user submits a query. The retrieved context is then supplied to the LLM for response generation.

2.3.3 Theme 3: Semantic Search, Case Retrieval and Workflow Automation

Semantic search represents legal documents using vector embeddings and retrieves information according to contextual similarity. This can help identify conceptually related legal materials even when the query and document use different terminology.

Legal Case Retrieval is another important research area. Effective retrieval cannot depend only on textual similarity because factors such as jurisdiction, court hierarchy, and legal authority can affect the relevance of a case. Consequently, legal retrieval systems require domain-specific evaluation in addition to general semantic similarity.

Recent LegalTech systems also explore AI-assisted workflow automation, including document organization, legal research, calendar management, hearing reminders, and case-record management. Nirnay-X combines these functions with document analysis and RAG-based legal assistance within a modular architecture.

2.4 Comparative Summary of Reviewed Works

Existing legal information systems can be broadly distinguished according to their retrieval and processing approaches.

Traditional legal databases provide access to authoritative legal materials but generally require users to formulate searches and manually interpret the retrieved information. Keyword-based systems can efficiently locate exact terms but may not capture semantic relationships between differently expressed legal concepts.

Machine Learning and NLP approaches automate specialized tasks such as classification, extraction, and summarization. LLM-based systems provide more flexible conversational interaction but may generate unsupported information when they are not connected to reliable external sources.

RAG-based systems combine retrieval and generation, providing a mechanism for supplying external legal information to an LLM. However, the quality of the final response remains dependent on the retrieval process and underlying knowledge base.

Approach	**Main Capability**	**Key Limitation**
Traditional legal databases	Access to legal sources	Manual search and interpretation
Keyword search	Exact-term retrieval	Limited semantic understanding
ML/NLP systems	Classification, extraction and summarization	Often task-specific
LLM-based systems	Conversational interaction and generation	Risk of unsupported outputs
RAG systems	Retrieval-supported generation	Dependent on retrieval quality and source coverage

Integrated LegalTech systems	Research and workflow support	Integration and domain-specific reliability remain challenges

2.5 Identification of Research Gap

The literature indicates that AI has been applied to legal document analysis, information retrieval, summarization, legal question answering, and case retrieval. However, the reviewed material identifies several areas requiring further integration.

First, legal information retrieval requires more than keyword matching because legally relevant concepts may be expressed using different terminology. Second, LLM-based assistance requires external legal information and verification because generated responses can contain errors. Third, the effectiveness of RAG depends on the quality, completeness, and currency of the retrieved knowledge base.

Another identified gap is the integration of multiple legal functions. Existing approaches often concentrate on individual activities such as legal search, contract analysis, or document summarization. The report proposes combining document analysis, semantic retrieval, conversational assistance, legal precedent retrieval, and workflow management.

The literature also indicates a need to consider non-specialist users. Legal AI systems can provide technical assistance to professionals, but presenting legal information in understandable language for citizens remains an important design consideration.

2.6 Formulation of the Research Proposal

Based on the identified gaps, Nirnay-X is proposed as an integrated AI-powered legal intelligence platform. The system combines legal document processing, vector-based semantic search, RAG, LLM-based interaction, legal information retrieval, and workflow-management features.

The proposed architecture follows a modular pipeline. A legal document is uploaded and processed; its text is divided into meaningful chunks, embeddings are generated and stored in a vector database, and relevant content is retrieved when a user submits a query. The retrieved information is then provided to the LLM as contextual input for generating a response.

The proposal is designed as a decision-support system rather than a replacement for legal professionals. Its evaluation therefore considers multiple dimensions, including document-processing performance, retrieval quality, response relevance and support, system performance, and functional reliability. The report specifically identifies Precision@K, Recall@K, and MRR as retrieval measures, alongside response-level evaluation and human assessment.

The research proposal consequently focuses on integrating established AI techniques into a single legal information and workflow-support platform while explicitly recognizing the limitations of automated legal assistance.

3. Societal Impact and SDG Alignment

3.1 Primary SDG Relevance

3.1.1 Mapped SDG and Target

The primary Sustainable Development Goal (SDG) relevant to Nirnay-X is **SDG 16: Peace, Justice, and Strong Institutions**. SDG 16 includes Target **16.3**, which calls for promoting the rule of law and ensuring equal access to justice. It also includes Target **16.10**, concerning public access to information in accordance with applicable laws and agreements.

Nirnay-X is designed as an AI-based legal information and decision-support platform. It processes legal documents, retrieves relevant legal information, provides simplified explanations, and supports legal research and case organization. These functions relate most directly to the information-access and justice dimensions of SDG 16.

3.1.2 Alignment Justification

Nirnay-X aligns with SDG 16 primarily through its focus on improving access to and understanding of legal information. The platform allows users to upload documents such as contracts, judgments, petitions, legal notices, and FIRs. The documents are processed, converted into embeddings, and indexed for semantic retrieval. When a user submits a query, relevant document fragments are retrieved and supplied to the language model through the RAG pipeline.

For non-specialist users, the system is intended to explain legal terminology and document content in simpler language. For legal professionals, it provides support for legal research, retrieval of related judgments, case organization, hearing schedules, and timelines.

This alignment should be interpreted as a **technological contribution to legal information access**, rather than evidence that the system itself achieves equal access to justice. The project does not provide legally binding advice, replace

advocates or judges, or make judicial decisions. Its stated role is decision support and information assistance.

3.2 Breadth of SDG Linkages

Nirnay-X has potential connections with additional SDGs, but these relationships are less direct than its connection with SDG 16. The secondary linkages should therefore be understood as potential contributions arising from the platform's use and functionality rather than as demonstrated social outcomes.

3.2.1 Secondary SDG 1: SDG 3 – Good Health and Well-being

Nirnay-X has an indirect relationship with **SDG 3: Good Health and Well-being**. SDG 3 focuses on ensuring healthy lives and promoting well-being, including access to health services and universal health coverage.

The connection to Nirnay-X arises because legal documents and legal procedures can sometimes relate to matters affecting individuals' access to services, rights, safety, and welfare. The platform can assist users in understanding the legal information contained in relevant documents and identifying potentially applicable legal provisions.

However, Nirnay-X is **not a healthcare system** and does not diagnose medical conditions, provide medical treatment, or measure health outcomes. Therefore, the project cannot by itself be considered a direct implementation of SDG 3. Its contribution is limited to the potential role of legal-information accessibility in situations where legal understanding is relevant to an individual's welfare.

3.2.2 Secondary SDG 2: SDG 9 – Industry, Innovation and Infrastructure

A stronger secondary linkage exists with **SDG 9: Industry, Innovation, and Infrastructure**, which includes fostering innovation and developing technological capabilities.

Nirnay-X applies AI, NLP, LLMs, RAG, vector databases, semantic search, and software-based workflow automation to a legal information-management problem. Its architecture combines document processing, embedding generation, semantic

retrieval, AI response generation, case analysis, and calendar-based workflow management.

The project, therefore, represents an application of current AI and information-retrieval technologies to LegalTech. Its contribution to SDG 9 is primarily technological: it demonstrates how established AI components can be integrated into a domain-specific software system. The report also identifies future possibilities such as integration with additional legal databases, multilingual assistance, mobile applications, cloud infrastructure, and enhanced security.

3.3 Stakeholder Analysis

Nirnay-X involves several stakeholder groups with different requirements and potential uses.

Legal professionals are a primary stakeholder group. Lawyers may use the system for document analysis, legal research, retrieval of related judgments, case organization, and hearing or deadline management. The platform is intended to reduce repetitive information-management activities while leaving legal interpretation and final decisions to qualified professionals.

Common citizens are another important stakeholder group. They may use the platform to understand legal documents, terminology, and potentially relevant legal provisions. The system is designed to provide explanations in more accessible language, but users should verify important legal matters with qualified professionals and authoritative sources.

Legal institutions and information providers are relevant because the quality of the system depends on the legal information available to its retrieval system. The report identifies source authenticity, completeness, currency, metadata accuracy, and privacy as important data-quality considerations.

Technology developers and administrators are responsible for maintaining the software, databases, AI components, authentication mechanisms, and security controls. Their role is important because uploaded legal documents may contain confidential or personally identifiable information.

Academic and research stakeholders may use the project as an example of integrating RAG, semantic retrieval, LLMs, and workflow automation in a domain-specific application. The project does not propose a new AI algorithm; its contribution is primarily the integration of established technologies into a unified LegalTech platform.

Overall, the strongest SDG alignment of Nirnay-X is with **SDG 16**, because the project's central purpose concerns legal information access and decision support. **SDG 9** provides a clear secondary technological linkage, while the relationship with **SDG 3 is indirect and should not be interpreted as evidence of direct health impact**.

4. Universal Human Values & Harmony with Nature

4.1 Impact on Material & Plant/Bio Orders

4.1.1 Material Order

Nirnay-X is primarily a software-based legal assistance platform. Its material requirements include computers, storage, networking infrastructure, and computing resources for document processing, vector storage, and AI-based response generation.

The system processes legal documents digitally, including PDFs, and converts their content into searchable representations. This supports digital document handling and reduces the need for repeated physical handling of documents within the system. However, the project does not measure paper consumption or quantify any environmental benefit; therefore, a specific reduction in resource consumption cannot be claimed.

The modular architecture separates document processing, embedding generation, retrieval, AI response generation, and workflow management. This provides a structured approach to managing computational operations and future system expansion.

4.1.2 Plant/Bio Order

Nirnay-X does not directly interact with plants, ecosystems, or other biological systems. Its connection with the plant/bio order is therefore indirect.

The platform supports digital processing and storage of legal documents, but digital infrastructure itself requires electricity, computing equipment, and other physical resources. Since the project does not include an environmental-impact assessment, no quantified positive or negative effect on the plant/bio order is claimed.

4.2 Impact on Animal & Human Orders

4.2.1 Animal Order

The system has no direct functionality related to animals or animal-resource management. Consequently, no specific impact on the animal order is established by the project.

Any broader environmental effects would depend on factors such as the energy consumption and infrastructure used to operate the system. These factors have not been quantitatively evaluated in the project.

4.2.2 Human Order

The human order is the most directly relevant aspect of Universal Human Values for Nirnay-X. The platform is designed for legal professionals and common citizens. Citizens can use it to understand legal terminology and documents, while legal professionals can use it for legal research, document analysis, case organization, and hearing or timeline management.

The system emphasizes **responsible use of AI and preservation of human decision-making**. Nirnay-X is intended as an information and decision-support system rather than a replacement for advocates or judicial authorities. Legal interpretation and final decisions remain the responsibility of qualified professionals and judicial authorities.

Transparency and reliability are also relevant. The RAG architecture retrieves relevant legal information before generating a response, and the proposed validation process checks whether generated answers are supported by retrieved information and whether unsupported claims are introduced.

Privacy is another important human-value consideration because legal documents may contain personally identifiable or confidential information. The project therefore identifies access control, appropriate document handling, protection of credentials, and responsible processing of sensitive information as necessary safeguards.

4.3 Overall Mutual Fulfillment

Nirnay-X can contribute to mutual fulfillment by aligning technological development with human requirements and responsible resource use. At the human level, it aims to improve access to legal information, simplify complex documents, and support legal professionals in organizing case-related information.

At the technological level, the project integrates document processing, semantic retrieval, RAG, LLMs, and workflow-management components into a single platform.

The project does not establish a measurable environmental impact, so environmental benefits should not be overstated. Its strongest connection with universal human values is instead through **human-centered assistance, responsible AI use, privacy, transparency, and respect for human judgment**.

5. Research Methodology

5.1 Research Approach

The research follows a **design-and-development approach** for developing and evaluating Nirnay-X, an AI-powered legal intelligence platform. The methodology includes problem identification, system design, data preparation, implementation, testing, and validation. The overall workflow consists of document processing, text chunking, embedding generation, vector storage, semantic retrieval, Retrieval-Augmented Generation (RAG), response generation, and case-management functions.

5.1.1 Justification of Chosen Approach

A modular approach was selected because Nirnay-X combines document processing, semantic retrieval, LLM-based generation, tool calling, and workflow management. Separating these components allows individual modules to be developed and evaluated independently. The methodology also separates retrieval from generation, making it possible to assess whether responses are supported by retrieved legal information.

5.2 Proposed System / Algorithm / Framework

5.2.1 System Architecture / Conceptual Model

Nirnay-X follows a pipeline-based architecture. Legal documents are uploaded, processed, divided into smaller chunks, converted into vector embeddings, and stored in a vector database. When a user submits a query, the query is embedded and compared with stored representations. Relevant document fragments are retrieved and provided to the LLM as context through RAG. The generated response is then presented to the user.

The platform also includes case analysis, related-case retrieval, hearing management, calendar integration, and case timeline functionality.

5.2.2 Algorithmic / Design Details

The principal workflow is:

1. Upload a legal PDF.
2. Extract and preprocess the text, using OCR where required.
3. Divide the text into meaningful chunks.
4. Generate embeddings and store them with metadata.
5. Convert the user's query into an embedding.
6. Perform semantic similarity search.
7. Retrieve relevant top-K document chunks.
8. Construct the context for the LLM.
9. Generate a response using RAG.
10. Display the response and retrieved information.

The document-processing and chunking stages prepare legal material for retrieval, while the RAG stage provides retrieved context to the language model before response generation.

5.2.3 Novelty of the Proposed Approach

The novelty of Nirnay-X primarily lies in the **integration of established technologies** rather than the creation of a new AI algorithm. The platform combines semantic search, RAG, AI-assisted document analysis, legal information retrieval, conversational interaction, and workflow management in one modular system. It is intended for both legal professionals and common citizens.

5.3 Data Collection and Analysis

5.3.1 Data Sources / Dataset Description

The system uses two broad categories of information: **user-provided legal documents** and **legal knowledge sources**. Documents may include contracts, judgments, petitions, notices, FIRs, constitutional provisions, statutory provisions, and case-related information.

Processed documents are divided into chunks and associated with metadata such as document name, type, section, page number, case information, date, and source where available.

The report emphasizes source authenticity, completeness, currency, metadata accuracy, duplicate handling, and privacy.

The report does **not specify a verified numerical dataset size**, so no dataset count is claimed here.

5.3.2 Analysis Techniques

The analysis involves:

- Text extraction and preprocessing
- Semantic chunking
- Vector embedding generation
- Similarity-based retrieval
- Top-K retrieval evaluation
- RAG response evaluation
- Functional testing
- Human evaluation and error analysis

The proposed evaluation queries cover legal provisions, constitutional questions, case-law retrieval, document-based questions, legal terminology, case-specific queries, and procedural questions.

5.4 Experimental Design and Validation Plan

Validation is performed at multiple levels rather than treating Nirnay-X as a single black-box system. Component validation covers document processing, text extraction, chunking, embeddings, vector storage, semantic retrieval, LLM integration, tool calling, and calendar management.

Retrieval validation compares retrieved results with manually identified relevant documents or passages. RAG validation examines the relevance of retrieved

context, whether the answer addresses the query, and whether important claims are supported by the retrieved information.

5.4.1 Evaluation Metrics

The proposed evaluation covers five areas:

1. **Document Processing:** text extraction accuracy and processing success rate.
2. **Information Retrieval:** Precision@K, Recall@K, and MRR.
3. **Response Quality:** faithfulness, answer relevance, and context relevance.
4. **System Performance:** response time and document-processing time.
5. **Functional Reliability:** successful execution of tools and workflow operations.

Human evaluation may additionally consider relevance, clarity, completeness, source support, appropriate uncertainty, and overall usefulness.

5.4.2 Baseline / Benchmark Comparison

The current report defines evaluation metrics and validation procedures but does **not provide a completed numerical comparison with a specific external benchmark or legal-AI system**. Therefore, no unsupported benchmark results are included.

A future controlled comparison could evaluate the proposed semantic retrieval approach against conventional keyword-based retrieval using the same legal queries and evaluation metrics.

5.5 Tools, Technologies and Resources

Nirnay-X combines technologies for AI, NLP, information retrieval, document processing, databases, backend development, and workflow management. Python is identified as the primary programming language for the backend and AI components.

The main components include:

- **Python** – backend and AI development.
- **LLM** – natural-language understanding and response generation.
- **RAG** – retrieval-supported response generation.
- **Embedding model** – vector representation of legal text and queries.
- **Vector database** – storage and semantic retrieval.
- **PDF/OCR processing** – extraction of machine-readable text.
- **Semantic search** – contextual information retrieval.
- **Tool calling** – interaction with structured application functions.
- **Calendar and timeline management** – organization of case-related events.

Overall, the methodology provides a structured connection between the research objectives, system design, data processing, retrieval, response generation, and validation of Nirnay-X

6. Expected Outcomes and Dissemination Plan

6.1 Expected Research Contributions

Nirnay-X is expected to contribute an integrated approach to AI-assisted legal information access and document analysis. The proposed platform combines legal document processing, semantic search, retrieval-augmented generation (RAG), large language models (LLMs), and workflow management functions within a single system.

The expected contributions are:

- **AI-assisted legal document analysis:** Support processing and understanding of contracts, judgments, petitions, notices, and FIRs.
- **Semantic legal information retrieval:** Retrieve relevant legal content based on contextual similarity rather than relying only on exact keywords.
- **RAG-based legal assistance:** Provide responses using retrieved legal information as contextual input to the language model.
- **Simplified legal information:** Present legal terminology and document content in a form that can be more accessible to users without legal expertise.
- **Integrated workflow support:** Provide case organization, hearing schedules, timelines, and related document-management functions for legal professionals.
- **Modular architecture:** Maintain separate components for document processing, embeddings, retrieval, response generation, and workflow management, allowing future extensions.

The project is intended as an **information and decision-support system**, not as a replacement for lawyers, judges, or other legal authorities. Therefore, its expected contribution is to support legal information access and research rather than independently provide legally binding decisions.

6.2 Mode of Dissemination

The findings and development work can be disseminated through academic, technical, and project-based channels. Dissemination should distinguish between what has been completed and what remains a future possibility.

6.2.1 Conference Publication

The work may be prepared as a conference paper focusing on the integration of RAG, semantic retrieval, legal document processing, and AI-assisted legal information systems.

A conference submission could present the system architecture, methodology, evaluation framework, implementation, and experimentally obtained results. Any performance claims included in such a publication should be supported by the project's measured evaluation results.

6.2.2 Journal Publication

A more detailed version of the research may be considered for journal publication. A journal manuscript could provide a more extensive discussion of the literature, system architecture, data methodology, evaluation methodology, limitations, and future scope.

Particular emphasis could be placed on the relationship between semantic retrieval and RAG-based response generation in a legal-information setting. The report identifies retrieval quality, response faithfulness, relevance, system performance, and functional reliability as separate evaluation dimensions.

At present, publication should be described as a **potential dissemination route**, not as an achieved publication.

6.2.3 Patent Potential

The project may be examined for intellectual-property potential, particularly in relation to the specific system architecture and integration of legal document processing, retrieval, RAG, and workflow-management functions.

However, the report does not establish that Nirnay-X currently satisfies the legal requirements for patentability. The use of established technologies such as LLMs, vector databases, semantic search, and RAG does not by itself establish patentability. A formal assessment would require examination of applicable intellectual-property rules, prior art, novelty, inventive step, and the specific implementation.

Therefore, **patent application should be treated as a possibility requiring further assessment**, rather than as a confirmed outcome.

6.3 Outcome Mapping Table

Expected Outcome	Research/Project Contribution	Possible Dissemination
Legal document analysis	AI-assisted processing and understanding of legal documents	Conference/Journal
Semantic retrieval	Context-based retrieval of relevant legal information	Conference/Journal
RAG-based assistance	Retrieval-supported natural-language responses	Conference/Journal
Simplified legal information	Improved accessibility of complex legal content	Project Demonstration/Publication
Case and workflow management	Integrated organization of legal activities	Technical Demonstration
Modular LegalTech architecture	Integration of AI and retrieval components	Conference/Journal
Future IP assessment	Examination of implementation-specific innovation	Patent Assessment

The expected outcomes are directly connected to the system capabilities described in the project. The platform processes documents, generates embeddings, performs semantic retrieval, and supplies retrieved information to the LLM through RAG.

6.4 Anticipated Challenges to Dissemination

Several challenges may affect dissemination of the work.

First, **legal-data availability and privacy** can restrict the use of real case documents. Legal documents may contain personally identifiable or confidential information, requiring appropriate access controls and data-protection measures.

Second, **evaluation reproducibility** may be affected by differences in legal datasets, retrieval configurations, language models, and knowledge-base coverage. For this reason, published results should clearly describe the evaluation dataset, retrieval configuration, metrics, and testing procedure.

Third, **AI reliability** remains an important consideration. Generated legal responses need to be evaluated against retrieved information because fluent language alone does not establish factual correctness. The proposed validation process therefore checks relevance, source support, and unsupported claims.

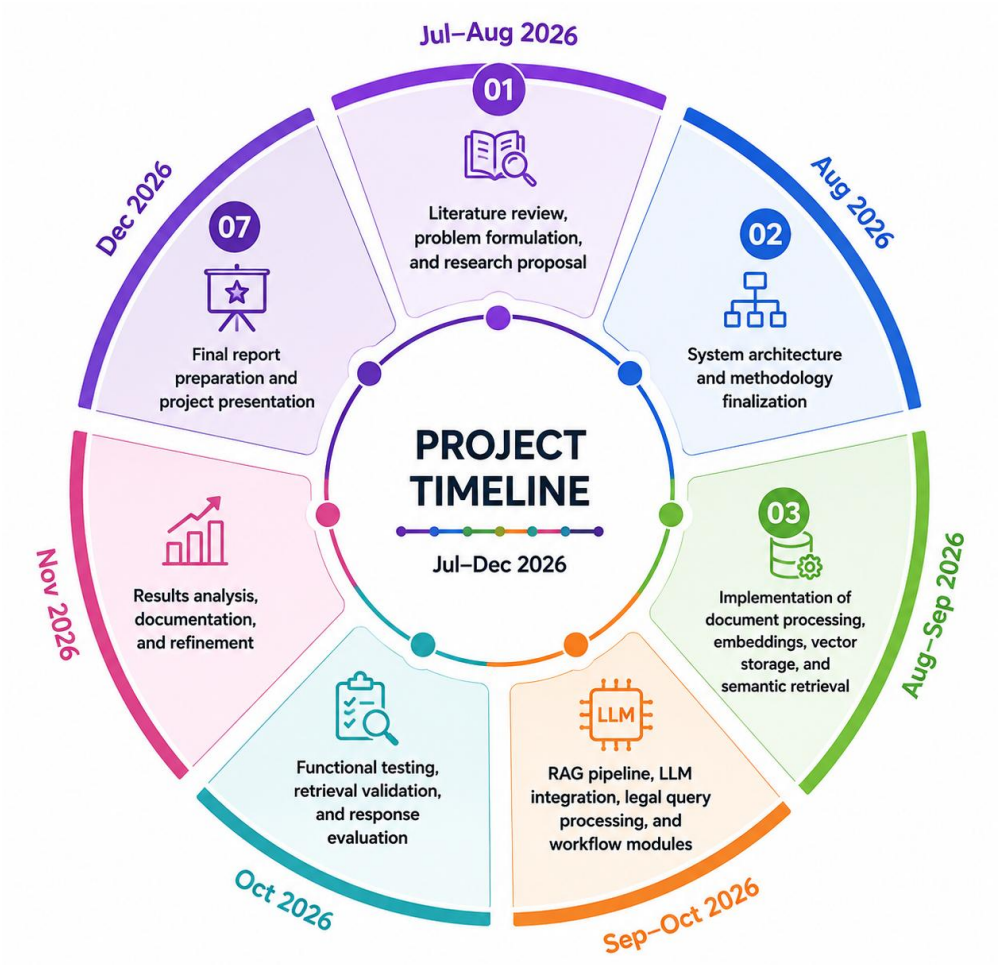
Finally, publication and patent dissemination may require additional work, including formal experimental results, documentation, prior-art analysis, and compliance with the requirements of the selected conference, journal, or intellectual-property authority.

Overall, Nirnay-X has potential for **academic publication, technical demonstration, and further intellectual-property assessment**, while these outcomes should be treated as future dissemination possibilities unless formally completed.

7. Project Plan and Professional Practices

7.1 Timeline and Milestones

The project plan for Nirnay-X is organized around the major stages of literature review, system design, implementation, testing, documentation, and final submission. The timeline is aligned with the project report and the 2026 project-development period.



The planned development follows the system workflow described in the report: document upload, text extraction, preprocessing, chunking, embedding generation, vector storage, semantic retrieval, RAG-based response generation, and supporting case-management functions.

7.1.1 Work Breakdown Structure

The project work is divided into the following activities:

1. **Literature and requirement analysis** – study of LegalTech, NLP, LLMs, semantic search, and RAG.
2. **System design** – architecture, database structure, module definition, and workflow design.
3. **Document-processing module** – PDF parsing, text extraction, preprocessing, and chunk generation.
4. **Retrieval pipeline** – embedding generation, vector storage, and semantic search.
5. **AI assistance** – RAG pipeline and LLM-based response generation.
6. **Legal and workflow modules** – case analysis, related-case retrieval, calendar, and timeline management.
7. **Testing and validation** – functional, retrieval, response-quality, and human evaluation.
8. **Documentation and presentation** – results, report preparation, and final demonstration.

The modular structure allows individual components such as PDF processing, embeddings, retrieval, LLM integration, and calendar functions to be tested separately.

7.2 Collaboration and Documentation

The project requires coordinated work among the team members for research, development, testing, documentation, and presentation. Project documentation should be maintained throughout development so that design decisions, implementation changes, testing procedures, and identified limitations can be recorded.

Version control should be used for source-code management, while shared cloud storage can be used for reports, datasets, documentation, and other project resources. Regular team discussions can be used to review progress, resolve implementation issues, and coordinate upcoming tasks.

Table 7.2: Team Roles and Responsibilities

Team Member	Primary Responsibility
Hridyansh Bhadauria	System development and implementation
Raghav Mittal	AI/ML and retrieval pipeline development
Ragini Goel	Data processing, testing, and validation
Rajat Pandey	Documentation, integration, and presentation

The four members are identified in the project report as the student team responsible for Nirnay-X.

7.2.1 Team Roles and Responsibilities

Although individual responsibilities may overlap, each member is assigned a primary area to improve coordination. Development tasks include document processing, embedding generation, semantic retrieval, RAG, LLM integration, and workflow functions. Testing activities include functional validation, retrieval evaluation, response assessment, and error analysis.

All members contribute to technical discussions, documentation, testing, and final presentation. This approach helps maintain shared understanding of the complete system rather than isolating knowledge within individual team members.

7.2.2 Collaborative Tools and Version Control

The project can use **Git and GitHub** for source-code version control and collaborative development. Shared cloud storage can be used for documents and

project resources, while collaborative document-editing tools can support report preparation.

Version control should maintain separate commits for major changes and preserve earlier working versions where necessary. Documentation should record important architectural decisions, testing procedures, known limitations, and changes to system components.

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